

PRIVATE NETWORK PROPOSAL

Own the network your lodge runs on.

A carrier-grade wireless build for Barefoot Addo Elephant Lodge — engineered to replace a rented, contended connection with infrastructure the lodge owns outright.

Prepared for

Barefoot Addo Elephant Lodge (Pty) Ltd

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01

Where things stand today

The lodge currently pays R18,105.60 a month for a shared, best-effort connection and a set of rented Wi-Fi access points. Nothing about that setup belongs to the lodge, and none of it scales without another monthly line item.

WHAT WE FOUND ON SITE

The backhaul radios that carry the whole property's connectivity are mounted on a leaning gum pole — not an engineered structure. It has held so far. It is not something to plan a lodge's connectivity around.

A GAP IN COVERAGE

The spa currently has no connectivity at all. It was never part of the original rollout, and extending the current rented setup to reach it would mean yet another monthly access-point fee stacked on top of an already contended network.

02

What we're building instead

A private wireless network on an engineered 9m mast, replacing the gum pole entirely. Two sector radios cover the full property from that single mast — the main lodge cluster at close range, and a second sector reaching out to the villas, the staff village, and the Director's House.

Twenty-eight access points go in across every building that needs coverage. That includes the spa — connected for the first time — alongside the boma, the school, the workshop, and every guest and staff area on the property.

The order of work matters. The private network — mast, cabinet, power, radios and access points — is built first. The Vodacom link is the final piece, and it can only go in once the mast is standing: Vodacom surveys the site against that mast before they will install.

BUILT FOR WHAT COMES NEXT

Because the sector coverage already reaches every corner of the property, adding CCTV later doesn't mean building new wireless infrastructure — it means plugging cameras into a network that's already live there. Security cameras at the boma, the villas, or the staff village become an endpoint cost, not an infrastructure project.

NOT A LOCAL FIX

CTTX is based in Port Elizabeth, but this build runs on Vodacom's national carrier network — over 14,000 towers across South Africa. If there's a second property, or a third, anywhere in the country, CTTX delivers and supports the same carrier-grade design there too. This isn't a one-off local install; it's a model that travels with the properties you own.

03

Why this has to work every time

A lodge network carries three different loads at once. This design was built to hold all three, not just the one that's easiest to sell.

CLIENT EXPERIENCE

Many guests are international, checking in with family across several time zones — sharing the day's sighting, confirming they've arrived safely, staying reachable somewhere their own cellphone has no signal. That only works if the connection is as reliable at 2am as it is at 2pm.

OPERATIONAL REQUIREMENT

Bookings, POS, staff communication and day-to-day lodge management all run over this same network. A dropped connection on a busy weekend isn't an inconvenience — it's a stalled payment or a missed booking.

SECURITY REQUIREMENT

Reserves deal with real incidents — a boundary breach, a security concern that surfaces after the office has closed for the night. When that happens, communication is what determines how fast a response reaches the right people. A network that goes quiet after hours is an operational risk, not a technicality.

99.9%

Designed availability

Built on an engineered mast in place of the gum pole it replaces, a Vodacom Business SLA link already confirmed for high availability at this site, and solar/battery power with no single point of failure. CTTX also monitors the network remotely, so an issue is caught before it becomes an outage.

04

The investment

One number, covering the complete build — design, hardware, mast, power system, installation, and commissioning.

TOTAL BUILD INVESTMENT

R355,000

Incl. VAT — paid once, owned outright

WIRELESS NETWORK

ePMP 4500L base station radios	×2
Sector antennas, 90/120°	×2
ePMP Force 4525L subscriber radios	×28
Wi-Fi 6 indoor access points	×28
Wi-Fi 6 outdoor access points	×2
Gigabit PoE+ managed switches	×4

MAST & FOUNDATION

9m galvanized steel lattice mast	×1
Concrete, sand & grit for the plinth	as req.
Reinforcing steel (rebar)	as req.
Foundation shuttering	as req.
Holding-down bolts	set
Mounting brackets & clamps	set

POWER SYSTEM

Weatherproof equipment cabinet	×1
MPPT solar charge controller	×1
Lithium battery bank & switchgear	×1
680W solar panels	×4
Grid/solar inverter	×1
Solar combiner & isolators	set

CABLING, EARTHING & INSTALL

Outdoor Ethernet & feeder cable	as req.
Armored power & solar DC cable	as req.
Copper earthing rods & bonding	set
Site survey & RF design	incl.
Installation, 2-technician team	incl.
Commissioning, training & handover	incl.

WHY THIS PRICE IS POSSIBLE

The mast and the equipment cabinet in this build are refurbished, CTTX-built units — engineered infrastructure we have on hand, offered here at a substantially reduced cost against buying new. Bought new, those two items alone would cost close to double. That is a meaningful part of how the total above is what it is, and it applies to this proposal specifically.

The financial case

Once built, the lodge's monthly bill drops and stays down — while the current provider's would only keep climbing.

Current monthly bill

R18,106/mo

New bill, once built

R8,949/mo

Payback period

1 yr, 1 mo

5-year ROI

169%



■ Continue with current provider (cumulative cost)

■ Build with CTTX (cumulative cost)

This proposal is offered on a 36-month agreement, with pricing locked in for that term. The comparison above shows what continuing to rent would cost over the same horizon, assuming a typical 8% annual CPI-linked escalation. The build cost is paid once, up front; the connectivity fee is the only recurring cost for the length of the agreement.

Next steps

Confirm the design

Line-of-sight for all 28 connection points is already confirmed by field survey. A final walk-through validates cable runs and exact mounting positions.

2 Approve the investment

Sign off on the R355,000 build and the 36-month connectivity agreement. The Vodacom feasibility submission goes in at the same time, so it runs in parallel with the build.

3 Build the private network

Mast, cabinet, power system, radios and access points go in first. This is the part CTTX builds and owns end to end, and it does not wait on Vodacom.

4 Vodacom site survey

Once the mast is standing, CTTX confirms the site is ready and Vodacom surveys against it. Their installation depends on being satisfied with that mast — which is why it has to be built first.

5 Carrier link installed

The last piece: the 100 Mbps symmetrical, uncapped, unshaped, unthrottled SLA link from Vodacom's Olifants Nek high site, installed in one go.

6 Go live

The current connection is decommissioned once the new network is tested and confirmed — including the spa, connected for the first time.

A NOTE ON TIMING

We have secured the wireless equipment for this build with our supplier. This kind of kit tightens up at this time of year, and we can't hold an allocation indefinitely — if it is sold on and the build stands over to next year, it will cost more. That is not a reason to decide in a hurry; it is simply the honest position on where the stock sits today.

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